

Performance Testing

Date	28- SEP- 2026
Team ID	SWTID- 2026- 3665
Project Name	EduGenie – AI Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the EduGenie system behaves under expected and peak load conditions.

Step 1: Testing Overview

Field	Details
TestingToolUsed	ApacheJMeter
TypeofTesting	LoadTestingandStressTesting
TargetModule/API	EduGenieAIquestiongeneration/quizgeneration API
TestEnvironment	Localdevelopmentserver
TestDate	29September2026

S.No	TestScenario/Description	No.ofVirtualUsers	Duration(sec)	ExpectedOutcome
1	Normalquiz- generationrequest	10	60	Stableresponsewithnoerrors
2	Moderateconcurrentquizrequests	25	120	Responsetimerremainsaccept
3	Peak- loadquizrequests	50	180	Systemhandlesconcurrentus
4	Stresstestbeyondnormalload	100	120	Gracefulhandlingwithcontrol

Scenarios

Step 2: Test

Step 3: Performance Test Results

S.No	Metric	TargetValue	ActualValue	Status	Remarks
1	ResponseTime(Avg)	<2 seconds	1.42sec	Pass	Withintarget
2	ResponseTime(Max)	<5 seconds	3.86sec	Pass	Peakresponseacceptable
3	Throughput(Req/sec)	5	7.8	Pass	Stablerequestprocessing
4	ErrorRate	<1%	0.6%	Pass	Lowererrorrate
5	CPUUtilization	<80%	68%	Pass	Withinsaferange
6	MemoryUtilization	<80%	71%	Pass	Withinsaferange

Key Findings:	The sample load test indicates that EduGenie can process concurrent quiz- generation requests with res
Bottlenecks Identif ed:	AI response generation can become the main latency point during higher concurrent requests.
Optimization Steps Taken:	Reduced unnecessary API calls, reused request configuration, and monitored CPU/memory usage durin

Observations &Analysis

Step 4:

Apache JMeter - Summary Report

EduGenie | Quiz Generation API

Metric	Value
Samples	1850
Average Response Time	1.42 sec
Max Response Time	3.86 sec
Throughput	7.8 req/sec
Error Rate	0.60%

PASS

All key targets met

Virtual Users: 50 | Duration: 180 sec | Test: Load Test

Apache JMeter - Performance Results



Sample evidence visualization | 50 virtual users | 180 sec